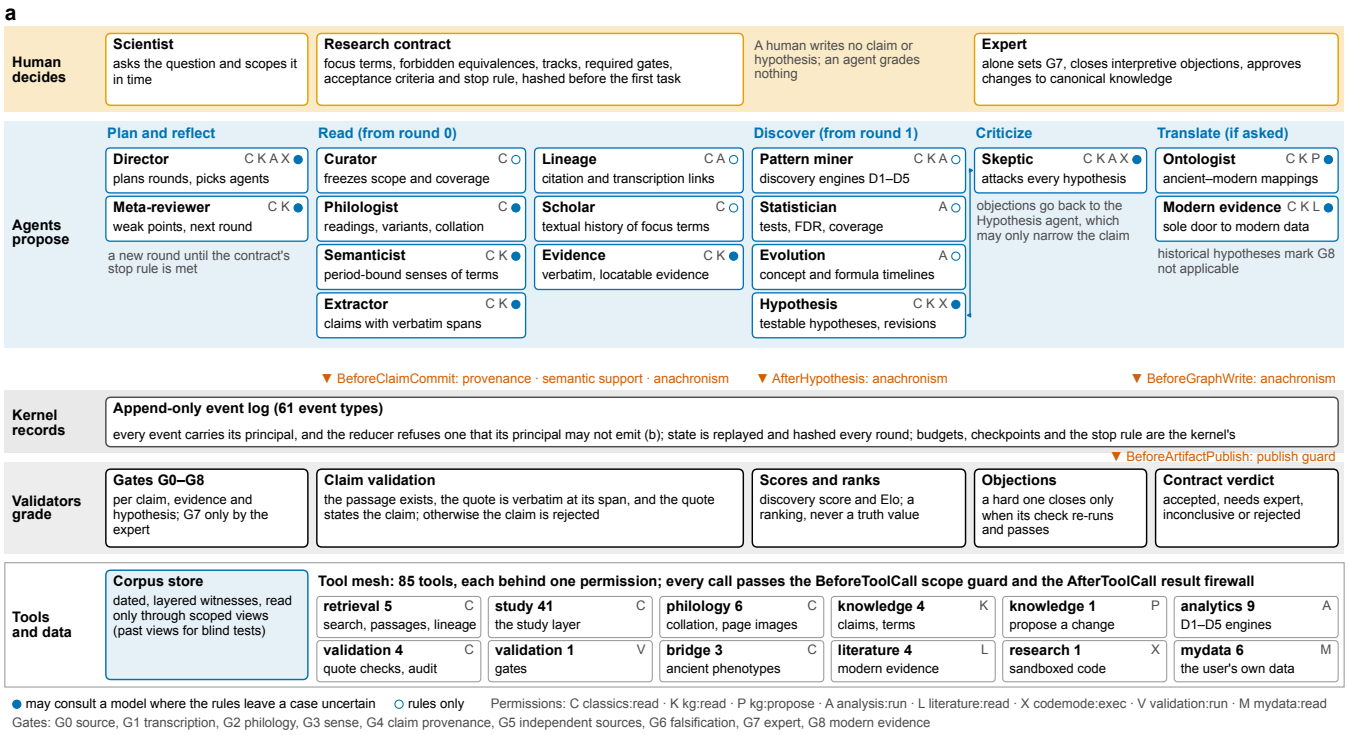
TaoChronos (道乃久): display items

Six figures, three Extended Data figures, two Extended Data tables and Supplementary Fig. 1, at their final print size (183 mm wide). Masters: paper/figures/out (PDF, SVG, 600 dpi PNG); legends: paper/legends.md; tables: paper/tables; Supplementary Information: paper/supplementary.



b

Events a principal may emit (count)	Agent	Validator	Human	Kernel
Plan and run tasks; scope the corpus; propose evidence, claims, lineage, observations; generate, review and revise hypotheses (21)	●	●	·	●
Call tools; record a decision; propose a change (7)	●	●	●	●
Call a model (1)	●	·	·	●
Grade: gates, scores, ranks, confirmation; evaluate the contract; publish (6)	·	●	·	●
Reject a hypothesis; retract evidence; write memory (3)	·	●	●	●
Close an objection; approve or reject a change (3)	·	●	●	·
Approve or reject a hypothesis (G7); steer a research (3)	·	·	●	·
Stop a research (1)	·	·	●	●
Open the contract, run rounds, spawn agents, route models, budget, checkpoint, complete (16)	·	·	·	●

Gate results: gates are graded by validators, never by the work they grade. Objections: an objection is closed by re-running its check (validator) or by an expert, never by a model. Only a human expert sets G7.

c

Jingui yaolüe ch. 13 on xiaoke 消渴 in three witnesses (punctuation removed)

omission (脱) 丸 → 圓

Deng Zhen print, 1340 男子消渴小便反多以飲一斗小便一斗腎氣丸主之
Wu Qian manuscript, 1395 男子消渴小便反多以飲一斗－－－－腎氣丸主之
Maijing, Yuan print, 1330 男子消渴小便反多以飲一斗小便一斗腎氣圓主之

圓 for 丸 avoids the name of Emperor Qinzong (桓, from 1126): that witness descends from an edition made after 1126

Claim	Shenqi wan 肾气丸 indicated for 消渴 with polyuria
Quote	男子消渴，小便反多，以飲一斗，小便一斗，腎氣丸主之
Passage	cm_jingui_dengzhen.00712
Locator	卷中 · 消渴小便利淋病脉證并治第十三 · p. 100
Witness	Deng Zhen print, 1340; Peking University Library
Work · layer	Jingui yaolüe, 200–219 · main text
Source · licence	CMETA, collated page by page · CC BY 4.0
Check in chat	✓ verbatim ≠ quote differs × no such passage

Fig. 1 | TaoChronos (道乃久): agents propose, validators grade, experts decide.

a, The harness as configured (web-canon profile), by principal (rows) and by the stage of a research round (columns). A scientist's question becomes a research contract, hashed before the first task. The Director plans each round and picks, from fifteen specialists, the agents the kernel spawns; each reads only through the research's scoped view and calls only the tools its permissions allow (letters). Filled circles, agents that may consult a language model where their rules leave a case uncertain (Fig. 3 called none). The Skeptic's objections return to the Hypothesis agent, which may only narrow a claim. Every write is an event in the log. Hooks (vermillion) refuse a tool call or result outside the research's scope, a claim whose quote is not verbatim or contradicts it, an equation of a historical with a modern concept, and a report on versions the contract never judged. Validators alone grade claims, gates G0–G6 and G8, scores and the contract's verdict, and close a hard objection only by re-running its check; the expert alone sets G7. **b**, Event authority: the principals allowed to emit each group of the 61 event types (number of types); the reducer refuses any other event. **c**, One clause of the *Jingui yaolüe* (chapter 13) in three witnesses, aligned character by character: the Deng Zhen print (1340), the Wu Qian manuscript (1395), which lacks 小便一斗, and the Yuan print of the *Maijing* (1330), which writes 圓 for 丸, a Song taboo form that dates its exemplar after 1126; shaded, variant positions. Below, the provenance chain of the first witness from claim to licence; the conversation client checks every cited passage (✓ verbatim, ≠ quotation differs, × no such passage). CMETA, 數位中醫校書郎. Agents, tools and permissions: Supplementary Tables 4–8.

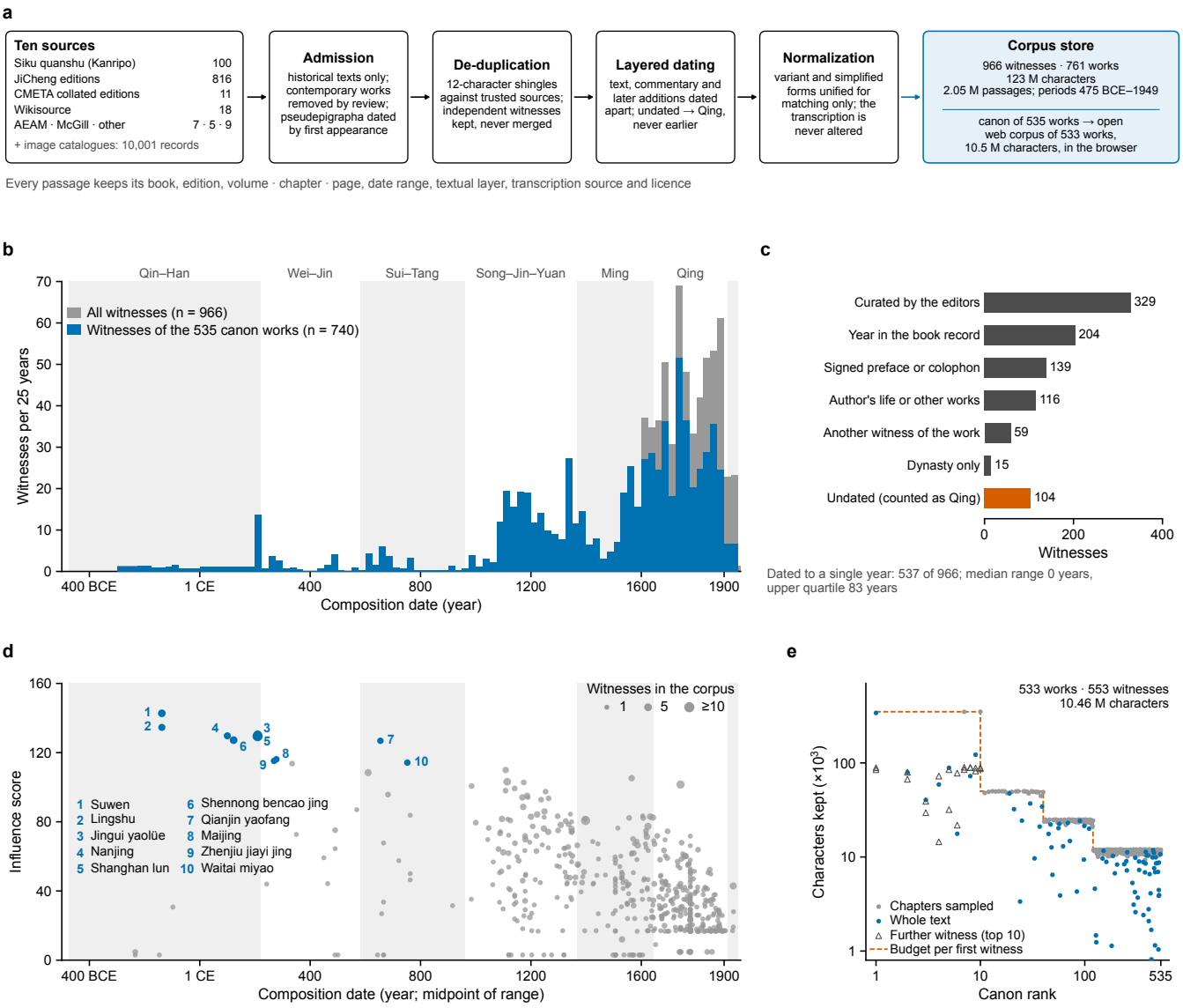


Fig. 2 | A dated, layered corpus spanning two millennia, and the canon its citations define.

a, Construction of the corpus. Transcriptions from ten sources pass admission (historical texts only), de-duplication against more trusted sources (normalized 12-character shingles; independently collated transcriptions are kept as separate witnesses), layer-aware dating (main text, commentaries, editors' notes and later additions dated apart; undated material counted as Qing, never earlier) and normalization used for matching only. **b**, Witnesses per 25-year bin by composition date; each witness counts once, its weight spread evenly over its date range (n = 966; blue, the 740 witnesses of the 535 canon works). Bands mark the analysis periods; the last, unlabelled band is the Republican period (1912–1949). **c**, Basis on which each witness was dated; the 104 undated witnesses are counted as Qing (vermillion), never earlier. **d**, Influence score of the 535 canon works against the midpoint of their composition range; point area, witnesses in the corpus (capped at 10). Score = 10 $\log_2$(1 + citing works not older than it) + 2 × periods in which it is cited + 2 $\log_2$(1 + citations) + 3 $\log_2$(1 + witnesses) + 6 × inclusion in the *Siku quanshu*; the ten highest (blue) are keyed by rank. Later works have had fewer chances to be cited: the score describes what the corpus cites, not a verdict on the tradition. **e**, Characters kept for each witness of the open web corpus against the canon rank of its work (log–log). Dashed line, the budget of a work's first witness (350,000 characters for ranks 1–10, then 50,000, 25,000 and 12,000); blue, texts kept whole; triangles, second and third witnesses of the ten highest-ranked works. Data: corpus catalogues (`corpus/catalog`) and the web corpus pack (`corpus/web`).

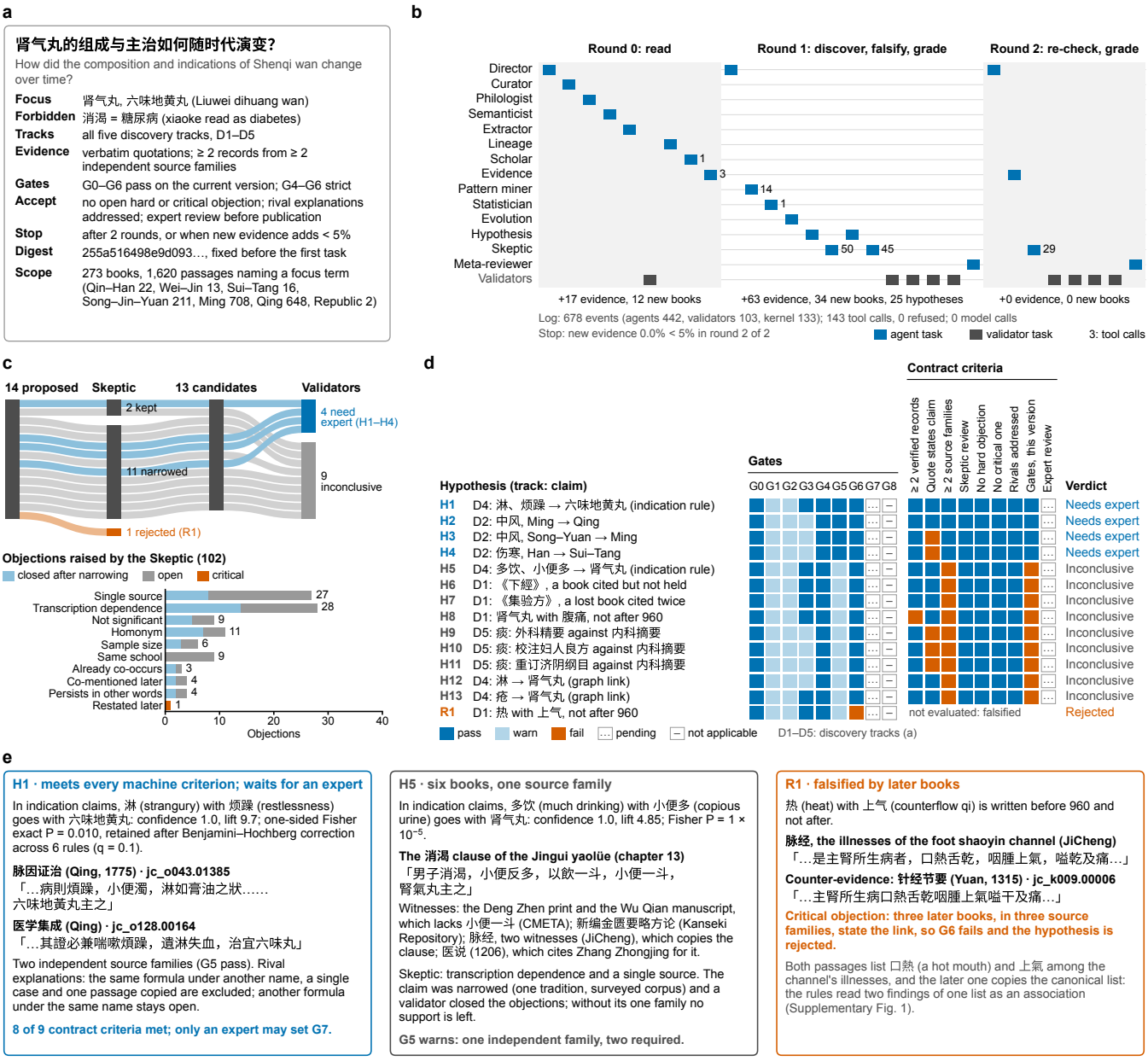


Fig. 3 | One research question, end to end.

a, The question, put to the harness on the open canon corpus (web-canon profile; rules only, no model called), and the contract fixed before the first task. **b**, The 30 tasks in the order the Director planned them over three rounds: agent tasks (blue), validator tasks (grey); numbers, tool calls. Below each round, the evidence records, new books and hypotheses it added; the run stopped after the last round the contract allowed, which added no evidence. **c**, The 14 hypotheses proposed in round 1: the Skeptic kept 2, sent 11 back to be narrowed and rejected 1; validators then judged the 13 candidates. Below, the Skeptic's 102 objections by check: closed by a validator once the claim was narrowed, left open, or critical. **d**, The final hypotheses against the gates and the contract's criteria. H1 and H2 meet every criterion a machine can check; H3 and H4 rest in part on a quotation that does not clearly state its claim, which the contract leaves to an expert. Nine fail on independent sources (G5): passage reuse, closed transitively, puts 36 of the 46 evidence books into one source family (Supplementary Fig. 1). **e**, Three outcomes with their verbatim evidence. H1 waits for an expert. H5 rests on six books of one family: the 消渴 clause of the *Jingui yaolüe*, its witnesses and copies. R1 is rejected because three later books state the link; they copy the canonical list of the kidney channel's illnesses that the earlier passage holds, and the rules read two findings of one list as an association. A second run writes the same report; report, event log, hypotheses and objections: Supplementary Data 1–2 and Supplementary Tables 9–10. D1–D5, discovery tracks: lost knowledge, concept evolution, formula evolution, hidden associations, contradictions.

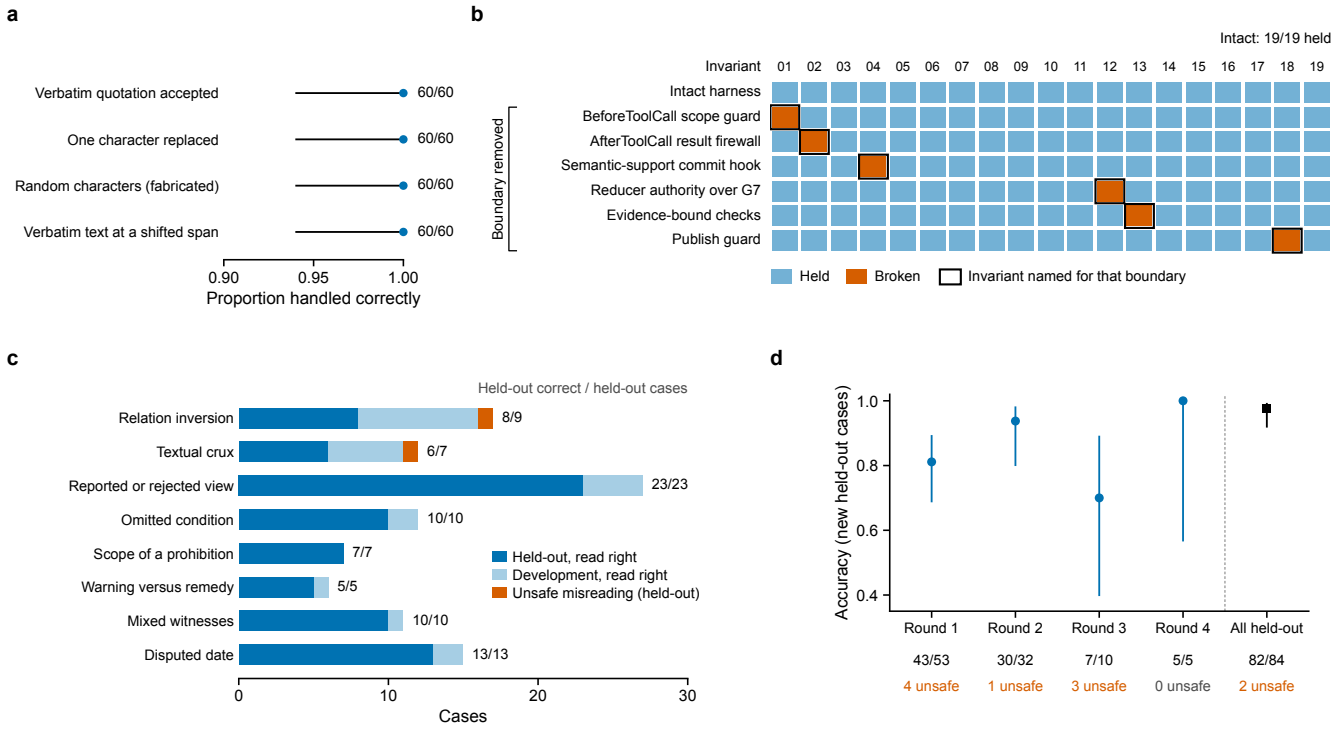


Fig. 4 | Quotations stay verbatim, the harness's boundaries hold, and hard readings stay safe.

a, Quotation checks on 60 random passages of the demonstration corpus: verbatim quotations accepted; quotations with one character replaced, of random characters, or verbatim at a shifted span rejected. Dots, proportions; lines, Wilson 95% confidence intervals. **b**, Nineteen invariants (I01–I19) run as probes on the intact harness and on six variants, each with one boundary removed by deleting its check at run time. Every removal broke the invariant named for its boundary (outlined) and no other; 127 of 133 cells held. In three cases a second safeguard limited the damage: without the scope guard the tool it no longer refused returned nothing of the withheld passage; forged G7 events reached the log but not the state; without the bound checks two of five impossible evidence records verified. **c**, 107 hard interpretive cases in eight classes, each a verbatim quotation with one correct reading (development, 23 cases used to write the rules; held-out, 84). Numbers, held-out cases read correctly. Both errors were unsafe, a reading that commentators dispute accepted as supported (the Dahuang mudan tang clause and the 里水 of the Yuebi jia zhu tang clause); they are reported, not patched. **d**, Accuracy in four blind rounds, each on new cases written before the rules that first read them, and on all held-out cases under the current rules. Lines, Wilson 95% confidence intervals; below, correct/total and unsafe errors. Data: `taochronos eval` on the demonstration corpus (**a**, **c**), `paper/figures/mutation_matrix.py` (**b**) and `docs/evals.md` (**d**).

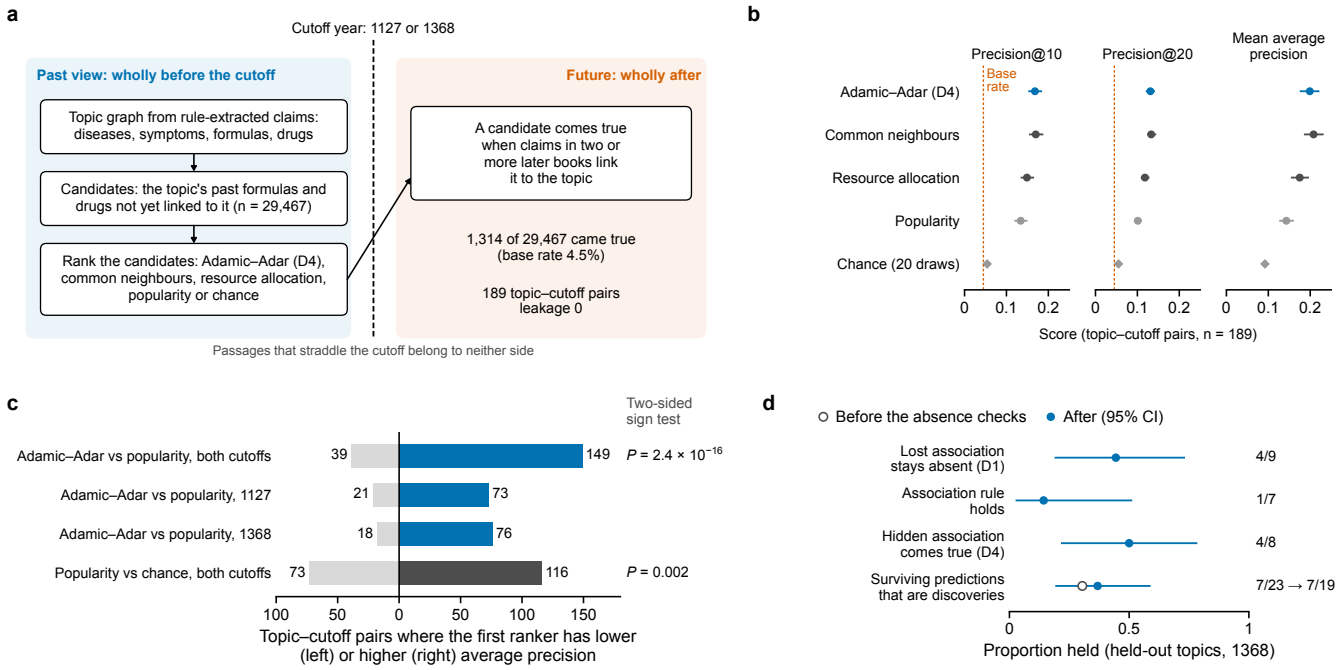


Fig. 5 | Rankings computed from the past anticipate the links later physicians wrote.

a, Blind temporal benchmark on the full corpus. At each cutoff (1127, 1368) the store splits into passages dated wholly before and wholly after it; passages that straddle it belong to neither side. For every disease or symptom of the lexicon with 150–6,000 past and at least 300 future passages, the formulas and drugs of its past graph not yet linked to it are ranked; a candidate comes true when claims in two or more later books link it to the topic. **b**, Precision at 10 and 20 and mean average precision of five rankers over the 189 topic–cutoff pairs in which at least one link came true. Dots, means; lines, 95% confidence intervals; chance, 20 random rankings; dashed line, base rate (4.5%). **c**, Paired comparisons of average precision over topic–cutoff pairs; two-sided sign tests, one tie excluded. **d**, Predictions of the whole pipeline run on the past view at 1368 for four held-out topics (胸痹, 霍乱, 虚劳, 痰饮): proportion borne out by later books, with Wilson 95% confidence intervals. Open circle, before the absence checks, which rejected two wrong 'lost' and two already attested 'hidden' associations and no discovery. No passage outside the past view was read (leakage 0). Values of the full-corpus run documented in `docs/evals.md`; the full store is not distributed, and Extended Data Fig. 2 repeats the benchmark on the open corpus.

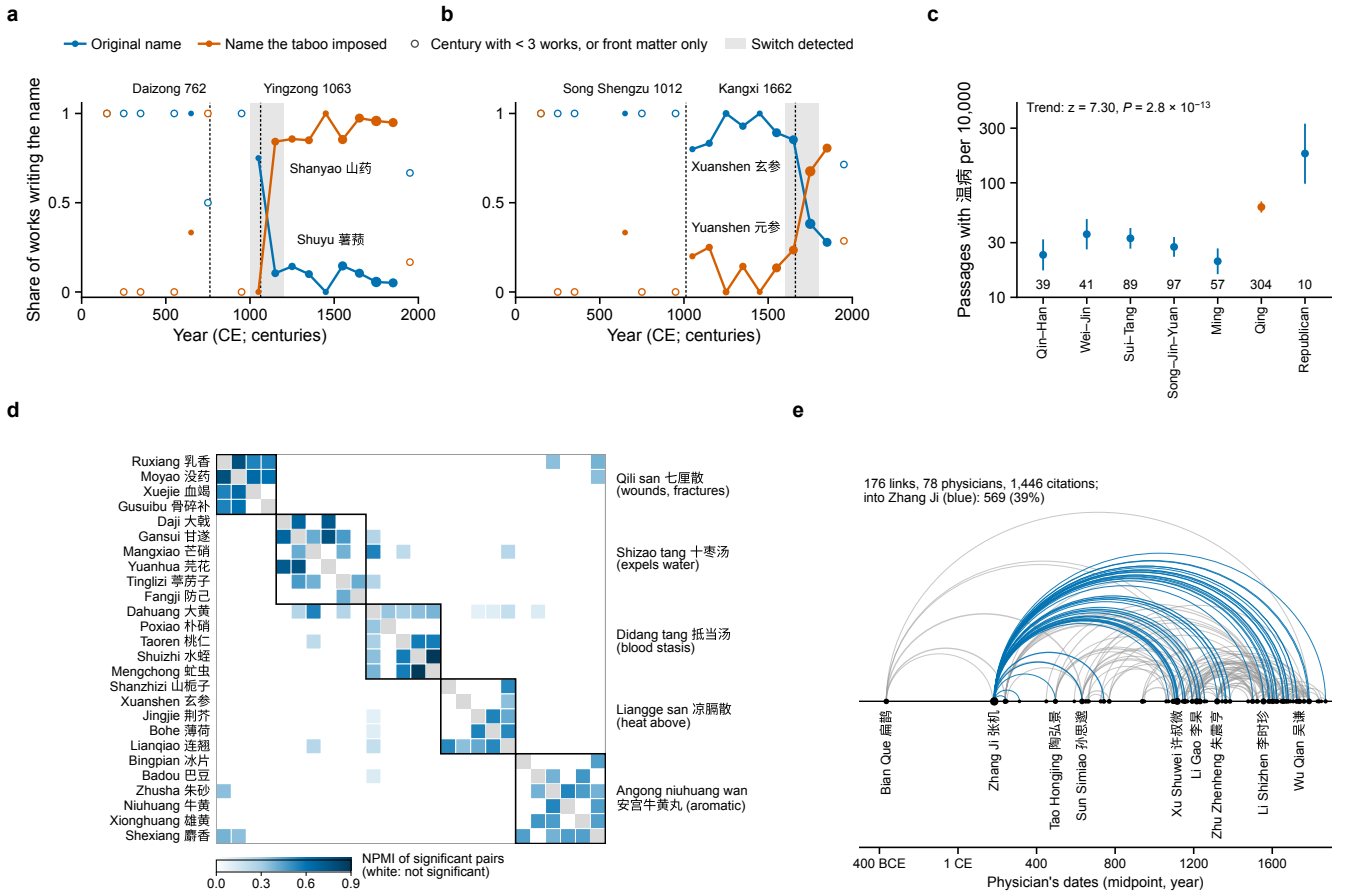
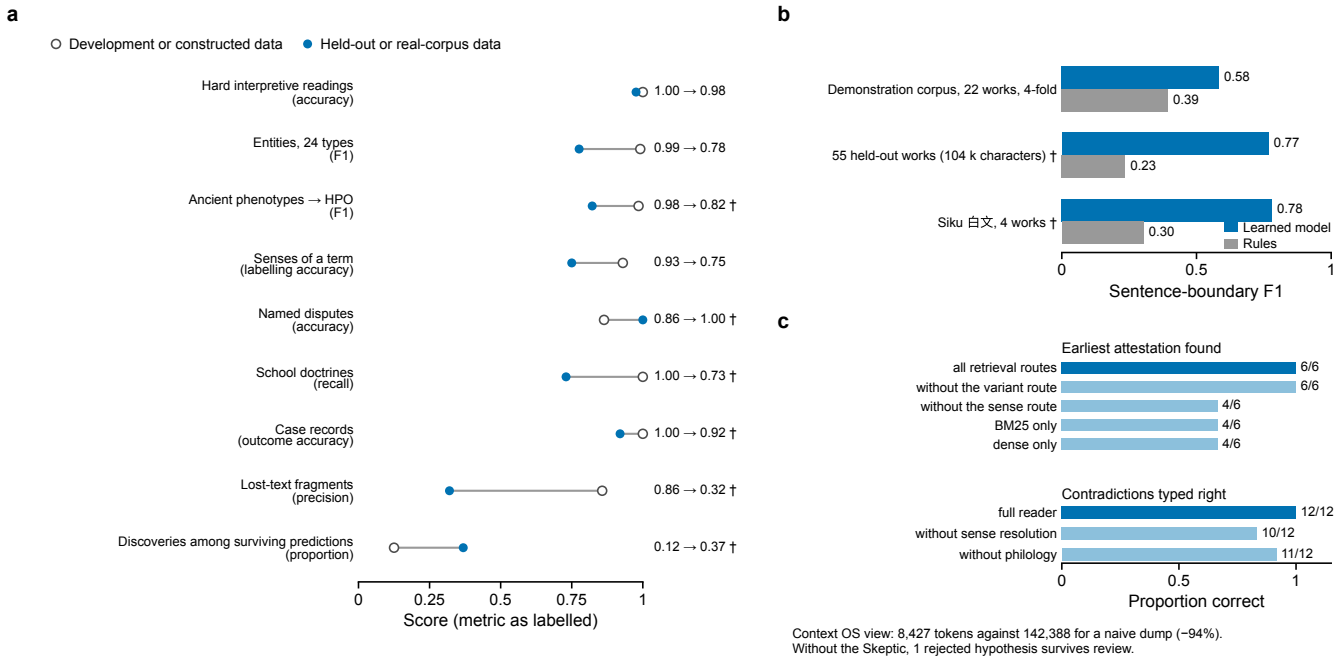


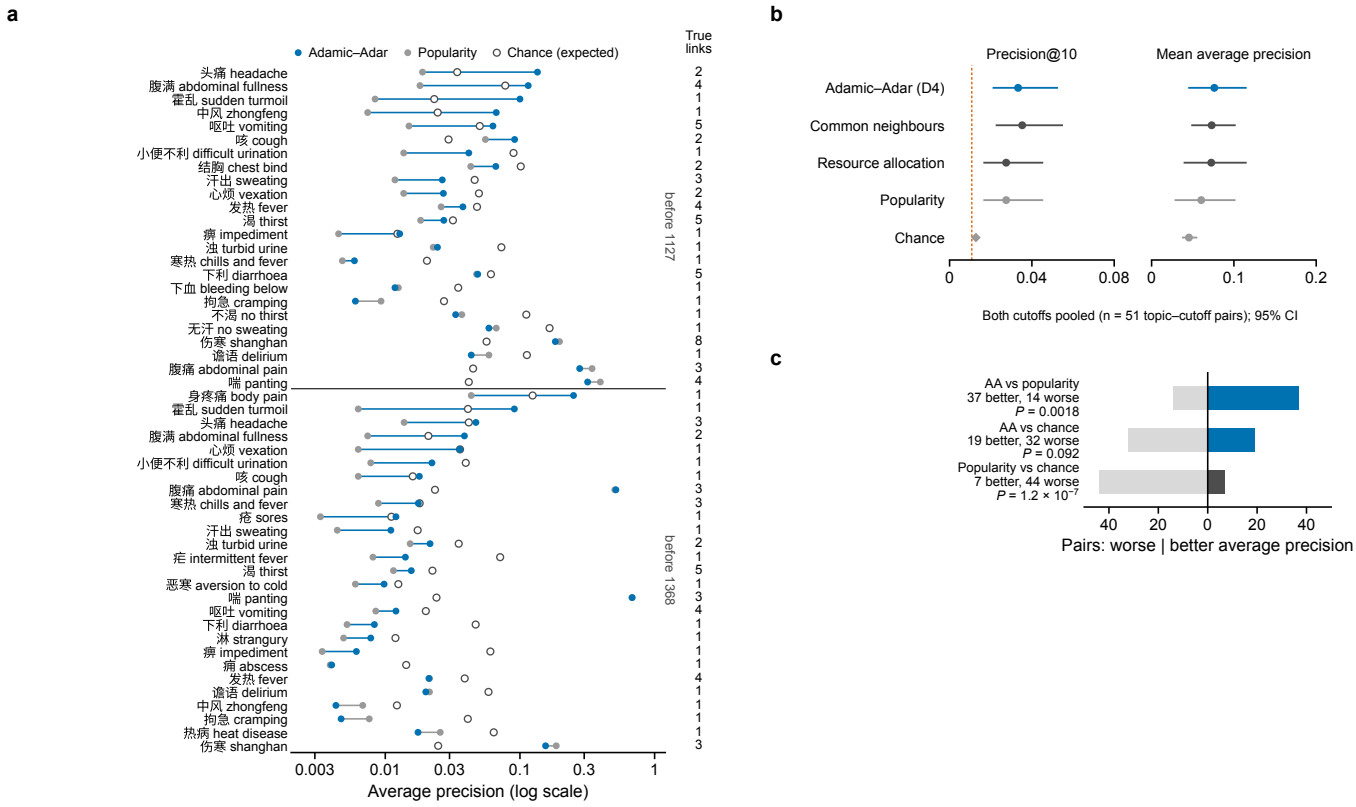
Fig. 6 | On the open canon corpus, the study layer recovers history written into the texts.

All panels are computed by the deterministic study layer on the open web corpus (533 works, 168,278 passages). **a**, **b**, Share of works per century writing an original drug name and the name a naming taboo imposed: 薯蓣 and 山药 (**a**), 玄参 and 元参 (**b**). Dashed lines, the years from which the names were avoided: Tang Daizong (李豫, 762) and Song Yingzong (赵曙, 1063) in **a**; the Song ancestor Shengzu (赵玄朗, 1012) and the Kangxi emperor (玄烨, 1662) in **b**; shading, the switch interval the tool detected; marker area, works in the century; open markers, centuries with fewer than three works or only undated front matter dated conservatively to 1911. A work writing both names counts for both. **c**, Passages mentioning 温病 (warm disease) per 10,000 passages by period, with Wilson 95% confidence intervals; numbers, passages with the term. Two-sided Cochran–Armitage test for trend across the ordered periods. **d**, Normalized pointwise mutual information (NPMI) of drug pairs co-occurring significantly in 4,554 compositions (a composition repeated in copies of one work counted once; pairs in at least ten works tested; one-sided Fisher exact test, Benjamini–Hochberg $q < 0.05$; 1,184 pairs). Groups are communities of the strong pairs (NPMI ≥ 0.3), found without formula names; shown (outlined) are the five of twelve whose drugs hold together most tightly (mean NPMI of their significant pairs). Each holds the core of a classical formula (right); liquorice (甘草), present in 43% of compositions, joins none. **e**, Citations between physicians: arcs join citing and cited physicians placed at the midpoints of their dates; width ∝ √citations; blue, citations of Zhang Ji (张机). Five links whose cited physician lived after the citing one (28 citations) are omitted as chronologically impossible.



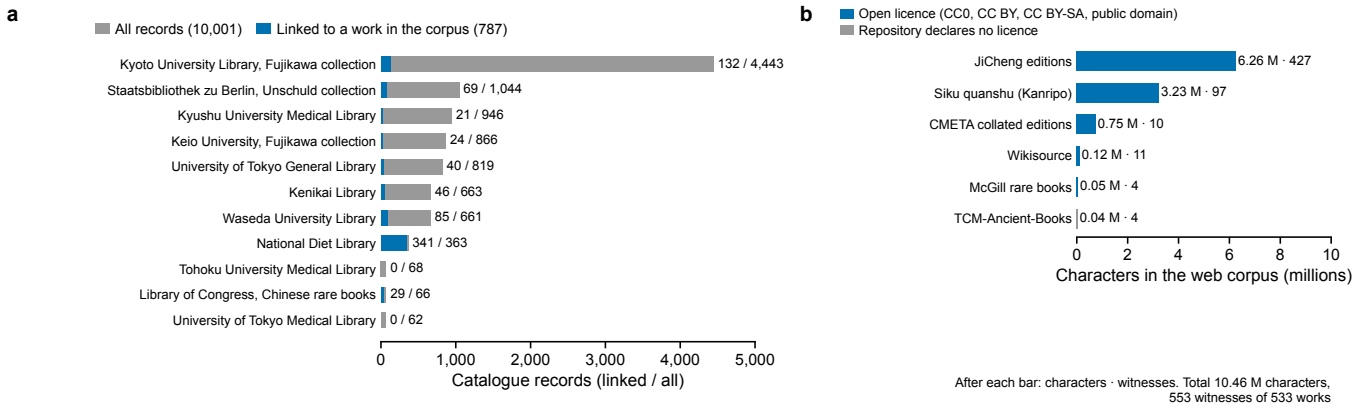
Extended Data Fig. 1 | Where the rules transfer and where they do not.

a, Development or constructed data (open circles) against held-out or real-corpus data (blue) for nine capabilities; numbers, the two scores; †, real-corpus value from the full corpus (`docs/evals.md`). Development sets were written with the rules they test. **b**, Sentence-boundary F1 of the learned punctuation model and of the rule baseline on the demonstration corpus (four folds), on 55 works held out of training, and on four unpunctuated *Siku* texts scored against their JiCheng punctuation (†, full corpus); every character was preserved in all settings. **c**, Ablations on the demonstration corpus: earliest-attestation accuracy with retrieval routes removed (6 terms), contradiction typing without sense resolution or philology (12 passage pairs), the context supplied to agents, and review without the Skeptic.



Extended Data Fig. 2 | The research-value benchmark on the open canon corpus.

The benchmark of Fig. 5 re-run with identical rules on the open web corpus (533 works, most sampled by chapters). **a**, Average precision of Adamic–Adar (blue), popularity (grey) and the expectation under chance (open circles) in each of the 51 topic–cutoff pairs in which at least one link came true (24 at 1127, 27 at 1368; log scale); lines join Adamic–Adar to popularity, blue where Adamic–Adar is higher; right, links that came true in the pair. More than half of the pairs have a single true link (28 of 51; 109 among 10,115 candidates), so one hit moves a ranking. **b**, Precision@10 and mean average precision of the five rankers, both cutoffs pooled (n = 51 pairs); dots, means; lines, 95% confidence intervals; dashed line, base rate (1.08%). **c**, Paired comparisons of average precision, two-sided sign tests. Adamic–Adar still ranks above popularity (37 of 51 pairs, P = 0.0018) but cannot be told from chance (19 better, 32 worse, P = 0.092), and popularity falls below chance (7 better, 44 worse, P = 1.2×10^{-7}). The benchmark needs the full corpus; the replication is reported as a limit of the open subset. Every pair and every prediction: Supplementary Tables 11 and 12.



Extended Data Fig. 3 | The witnesses behind the texts.

a, Catalogue records of printed and manuscript copies in eleven digitized collections (title, date, print or manuscript, shelfmark, IIIF manifest and licence; no image is downloaded) and those linked by title to a work of the corpus (787 records of 179 works); links made by title need checking by hand. **b**, Sources of the open web corpus in millions of characters and witnesses. Blue, openly licensed transcriptions: JiCheng, whose selection, arrangement, punctuation and notes on public-domain texts are released under CC0 1.0 and its own writing under CC BY 4.0 (the modern editors' part of the few texts it took from modern reconstructions of lost books excepted); CMETA, CC BY 4.0; Kanseki Repository and Wikisource, CC BY-SA 4.0; McGill, public domain. Grey, the four TCM-Ancient-Books texts that match no JiCheng text, whose repository declares no licence, carried on the decision of the releasing institute (IMPF-AI). Sources whose terms forbid redistribution are never included. IIIF, International Image Interoperability Framework.

Extended Data Table 1 | Sources of the corpus

Source	Licence	Works admitted	Characters (millions)	Transcription and notes
Kanseki Repository KR3e, Siku quanshu medical section (漢籍リポジトリ)	CC BY-SA 4.0	100	25.66*	Wenyuange Siku and Sibu congkan facsimiles; unpunctuated; catalogue with authors, dates, layers
JiCheng 笈成 edited texts, v1.4.8	CC0 1.0 (editorial work); base texts public domain§	816	93.64*	Volunteer transcriptions with modern punctuation; 41 of 857 works excluded
CMETA 數位中醫校書郎 collated editions	CC BY 4.0 (collation); base texts public domain	11	1.49	Collated page by page against images; passages carry page numbers and image links
Tōa Igaku Kyōkai 東亜医学協会 texts	© the society; local research only†	7	0.33	Suwen, Lingshu, Nanjing, Shanghan lun, Jingui yaolüe, Shennong bencao jing, Bian Que biography
Wikisource (Category 中醫, 醫書)	CC BY-SA 4.0	18	0.77	Works new after de-duplication of 627 candidates
McGill University Library rare books	Public domain	5	0.40	Qing prints and manuscripts on gynaecology
TCM-Ancient-Books; tcmoc	Not declared by the repositories ‖	8	0.37	Mostly simplified copies of JiCheng texts, de-duplicated work by work
Hugging Face classical-tcm-canon	Proprietary-commercial; local research only†	1	0.31	One work (Zhang Zhicong's commentary on the Shanghan lun)
Kanseki Repository catalogue (KR-Catalog)	CC BY-SA 4.0	–	–	Roles and life dates of 132 persons responsible for the Siku works
Image catalogues of 11 collections‡	Recorded per collection	–	–	10,001 records (787 linked to 179 works): title, date, print or manuscript, shelfmark, IIIF manifest
Total		966	≈123	2.05 M passages; 761 distinct works; analysis periods 475 BCE–1949

*Counted in the corpus store (README); the catalogues of these two sources do not record characters. Other counts sum the catalogue records of the admitted works.

†Used for local research only and never redistributed; the open web corpus carries none of the texts whose terms forbid redistribution.

§JiCheng releases its selection, arrangement, punctuation and notes on public-domain texts under CC0 1.0 and its own writing under CC BY 4.0 (https://jicheng.tw/tcm/copyright.html, consulted 29 September 2026); the open web corpus carries 427 JiCheng witnesses under these terms, crediting JiCheng. The modern editors' part of the few texts JiCheng transcribed from modern reconstructions of lost books is not covered.

‖ Mostly simplified-character copies of JiCheng texts; de-duplication removes every copy, and the 8 texts kept match no JiCheng text. The 4 in the open web corpus are carried on the decision of the releasing institute (IMPF-AI); the others are used for local research only.

‡Kyoto University (Fujikawa), Staatsbibliothek zu Berlin (Unschuld), Kyushu University, Keio University (Fujikawa), University of Tokyo (two libraries), Kenikai, Waseda University, National Diet Library, Tohoku University and Library of Congress. Only metadata are held; no image is downloaded.

Sources are ranked by trust for de-duplication (Kanripo, JiCheng, McGill, CMETA, AEAM, Wikisource, web collections); a candidate sharing enough normalized 12-character shingles with an admitted work is recorded as its copy, while McGill, CMETA and AEAM transcriptions are kept as independent witnesses. IIIF, International Image Interoperability Framework.

Extended Data Table 2 | Evaluation of TaoChronos across its three layers

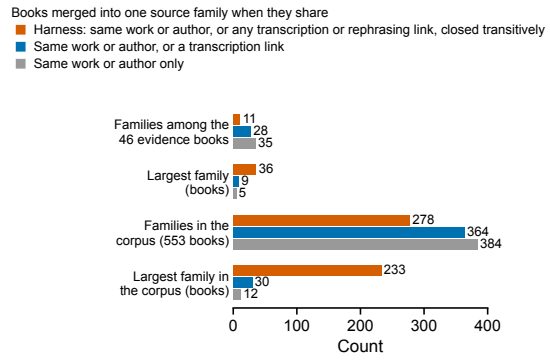
Layer	Suite	Data	Metric	Development or constructed	Held-out or real corpus
Boundaries	Invariants	19 probes of the harness under test	Held	19/19	Each of 6 boundaries removed breaks its invariant
Readings	Hard interpretive cases	107 cases in 8 classes	Accuracy (unsafe errors)	23/23 (0)	82/84 = 0.976 (2)
Research value†	Link anticipation	189 topic–cutoff pairs; 29,467 candidates	Precision@10	–	0.168 (popularity 0.134; chance 0.054); higher average precision than popularity in 149 of 189, $P = 2.4 \times 10^{-16}$
	End-to-end predictions	Held-out topics, cutoff 1368	Discoveries among survivors	1/8	7/19 = 0.37; leakage 0
Verification	Fabricated quotations	60 quotations × 4 probes	Handled correctly	–	60/60 in every probe; variant spellings accepted
	Provenance	Committed claims and evidence	Verbatim	–	100%; mean chain depth 7.8 levels
	Claim extraction	42 gold hyperedges	Precision / recall	1.00 / 0.976	–
	Temporal and anachronism guards	Constraints, earliest attestations, equations	Correct	6/6 · 6/6 · 10/10	Leakage 0; agent-asserted equivalence blocked
Deep discovery	Collation	5 traditions copied down a known stemma	Variant-unit recall / precision	0.998 / 0.999	–
	Semantic reuse	33 typed pairs	Accuracy	0.97	–
	Stratigraphy	5 composites; the Suwen	Layer accuracy	0.955	Yunqi chapters 7/7 in the minor layer†
	Lost-text fragments	Constructed lost work; Qianjin yaofang	Precision	0.86	0.32†
Task specification	Punctuation	55 held-out works (104 k characters)†	Boundary F1	–	0.767 (rules 0.235); characters kept 1.000
	Drug pairs	66,196 compositions†	Textbook pairs significant	–	7/7, lift > 1.5
V2	Entities (24 types)	41 development, 25 held-out sentences	F1	0.991	0.776
	Ancient phenotypes → HPO	20 development cases; 12 random passages†	F1	0.985	0.822
	Evidence audit	19 claims	Verdicts right (false SUPPORTED)	19/19 (0)	–

Development and constructed gold sets were written by the authors with the rules they test; they are regression tests of intended behaviour, not unbiased benchmarks. Only held-out and real-corpus columns estimate generalization.

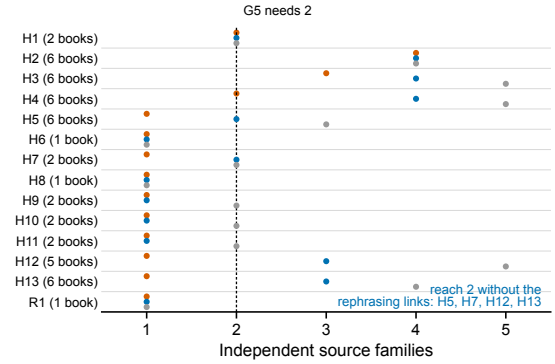
†Needs the full corpus store, which is not distributed; values from docs/evals.md. All other values were reproduced for this paper by `taochronos eval` on the demonstration corpus (40 suites, 47 s).

P value from a two-sided sign test of average precision over topic–cutoff pairs. HPO, Human Phenotype Ontology.

a



b



c

Failure	Where	What it did	What would catch it
Source families over-merged	H5, H7, H12, H13 (Fig. 3d)	the transitive closure chains 233 of 553 books into one family; with transcription links only, the four would rest on two or three families, but H5 and H13 rest on copies of one Jingui clause each, so one family is right for them	judge independence per claim, from the passages that support it, not per book
Co-listing read as association	R1; H8	R1: 口熱 and 上氣 are two items of the kidney channel's list of illnesses, which later books copy; H8: 小腹痛 belongs to the 尺脈伏 sentence, which gives 大平胃丸, not to 腎气丸	require the relation to be stated (indication, cause), not co-mention, before a lost or restated link counts
Concept drift on loose evidence	H2–H4	the evidence includes passages that do not name 中风 (迎風, 風氣百疾) and one Shanghan lun clause (傷寒中風) in four witnesses; the drifts' own permutation tests fail (P = 0.66, 0.89, 0.98), yet narrowing to 'exploratory' closes the objection and all three reach the expert	take drift evidence only from passages naming the term, copies once; make a failed test of the hypothesis's own statistic a hard objection
Case records read as doctrine	H9–H11	the 内科摘要 sentence and the 外科精要 passage are Xue Ji's records of treatments that failed (不應; 愈甚), not views; the 外科精要 and 校注妇人良方 witnesses are Xue Ji's editions (1547), whose notes and cases take the dates of the base texts (1263, 1248), so H9 sets Xue Ji against himself	read case records as records of practice; date editors' notes and cases as a later layer
Overlapping hypotheses	H2–H4; H9–H11	three drifts share the four witnesses of one clause, and three contradictions share one 内科摘要 passage	group hypotheses whose evidence overlaps before they are reviewed
Coverage of the open corpus	sources dossier	the open corpus keeps 8 of 24 chapters of 小儿药证直诀 (1119), none with 地黄丸, so the earliest composition of 六味地黄丸 found is of 1564; the 地黄丸 of 978–992 is another formula of the same name	state the sampled share beside every earliest attestation; rerun on the full store

Also reported, not repaired: the two unsafe readings of the held-out interpretive cases (Fig. 4c) and the open-corpus benchmark, too small to separate the ranking from chance (its replication of Fig. 5).

Supplementary Fig. 1 | Where the end-to-end run of Fig. 3 went wrong, and why.

a, Books merged into one source family under three rules: the harness's (the same work or author, or any transcription or rephrasing link between their passages, closed transitively), transcription links only, and the same work or author only. Under the harness's rule 36 of the 46 books that supplied evidence, and 233 of the 553 books of the open corpus, fall into one family. **b**, The independent families each final hypothesis rests on under the three rules; dashed line, the two that G5 requires. The hypotheses that G5 failed and that rest on two to three families once rephrasing links are left out (H5, H7, H12, H13) include two (H5, H13) whose evidence is copies of one clause, for which one family is right. **c**, The failure modes found by reading every surviving hypothesis against its evidence, with what each did and the check that would catch it. None is repaired here: the run is reported as it happened.